

Option A at $a = 5.00 \mu\text{m}$ with realistic lifetimes

Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$ Förster pair, ARC-computed BBR lifetimes, $\Omega_R = 2.9 \Omega_p$, bare Ω_p with $\sigma = 0.001771$ (Option 1 pulse)

TriQG examples/Average_fidelity_Vcc_newenergy · compiled 2026-04-11

TL;DR – Option A is worse than the paper baseline, but not for the reason you would expect. Four changes were stacked together in this run:

- (a) **Rydberg level swap** Rb $69D_{\frac{5}{2}} \rightarrow 66D_{\frac{5}{2}}$ and Cs $79D_{\frac{5}{2}} \rightarrow 76D_{\frac{3}{2}}$, with $\tilde{C}_3$ and C_6 updated to the Ireland et al. 2024 Table I values for this pair.
- (b) **Lifetimes** replaced by the ARC 3.10.2 `getStateLifetime` results at $T = 300$ K (see §3), which are **about half** of the values quoted in `main.tex` for the paper pair.
- (c) $\Omega_R/\Omega_p : 3.5 \rightarrow 2.9$ (user request).
- (d) **Option 1 pulse cleanup**: remove the ad-hoc 1.039975 multiplier on Ω_p and rescale $\sigma : 0.0014 \rightarrow 0.001771$ so the effective two-photon pulse area stays fixed at $\frac{\pi}{4}$. The pulse shape shifts by at most ≈ 4 MHz in peak height and is slightly smoother at the edges.

Result:

- OR gate: $\overline{F}_{\text{OR}} = 0.995584 \rightarrow \mathbf{0.991858}$, infidelity $4.42 \times 10^{-3} \rightarrow \mathbf{8.14 \times 10^{-3}}$.
- CCX gate: $\overline{F}_{\text{CCX}} = 0.997273 \rightarrow \mathbf{0.996620}$, infidelity $2.73 \times 10^{-3} \rightarrow \mathbf{3.38 \times 10^{-3}}$ (CCX does not use Ω_p / Ω_R and was not re-run under change (d); its value is from the previous Option A run).

What caused the drop: changes (b) and (c) together swamp the Option A level-swap gain. Decomposed:

- (b) **ARC lifetimes** add $\approx 10^{-3}$ infidelity per branch across both gates because T_1 is roughly halved.
- (c) $\Omega_R/\Omega_p = 2.9$ re-tunes the EIT dark state: the $|0, 0, * \rangle$ branch – which was flat at 0.9992 under the 3.5 value – drops to 0.9936 and grows a 0.0030 residual $|R \rangle$ population (15 $\times$ larger than before). This is the **biggest single contributor** to the new infidelity, and it is a pulse-parameter choice, not physics.
- (d) **Option 1 pulse cleanup** gives a small positive effect: the smoother edges improve the adiabaticity of the EIT dark state, cutting the $|0, 0, * \rangle$ residual $|R \rangle$ population from 0.0038 (fudged pulse) to 0.0030 ($\approx -21\%$), and lifting $\overline{F}_{\text{OR}}$ by $+2.6 \times 10^{-4}$ relative to the same Option A run with the fudged pulse.
- (a) **Level swap alone** would have given $\overline{F}_{\text{OR}} \approx 0.9956$ (slightly better than the paper) and $\overline{F}_{\text{CCX}} \approx 0.9973$ (essentially tied with the paper), matching the earlier prediction.

If you want a clean read of the Option A gain, re-run without (c): revert `omega_R_amp` to 3.5 * `omega_p_amp` and leave everything else (including the Option 1 pulse cleanup) as is.

1 Scope of this run

This run is a three-way parameter update to the Option A pair from the previous conversation, executed in the `_newenergy` folder:

1. **Physically swap the Rydberg pair** from the paper’s worked example (Rb $69D_{\frac{5}{2}}$ + Cs $79D_{\frac{5}{2}}$) to the Ireland, Pritchard, Shaffer, PRR 6, 013293 (2024) Table I row 1 candidate: Rb $66D_{\frac{5}{2}}$ + Cs $76D_{\frac{3}{2}}$. Both the $\tilde{C}_3$ dipole–dipole coefficient and the C_6^{CsCs} van der Waals coefficient are updated to the tabulated values for this new pair.

2. Use **ARC-computed lifetimes** for the two new Rydberg states at $T = 300$ K, replacing the earlier rough estimates we used two reports ago.
3. **Switch the pulse ratio** Ω_R/Ω_p from the paper’s 3.5 to 2.9 (requested by the user).
4. **Option 1 pulse cleanup**: remove the ad-hoc 1.039975 multiplier on Ω_p and slightly enlarge the super-Gaussian width $\sigma : 0.0014 \rightarrow 0.001771$ so the effective two-photon pulse area $\int \Omega_p^2/(2\Delta) dt$ stays fixed at its original design value of $\frac{\pi}{4}$. The pulse shape is visually indistinguishable from the previous one (peak difference $\lesssim 4$ MHz) and the total gate time is unchanged at 320 ns, but the smoother edges slightly improve the EIT-dark-state adiabaticity on the no-blockade branch.

Everything else stays at the paper-consistent values (lattice $a = 5.00$ μm , $\Omega_c = \Omega_p = 2\pi \times 50$ MHz, $\Omega_{cc} = 2\pi \times 100$ MHz, $\Omega_t = 2\pi \times 50$ MHz, $\Delta = 2\pi \times 500$ MHz, timing windows unchanged).

The previous `newenergy` run ($a = 5.25$ μm , V_{cc} overridden to +5 MHz, old lifetimes, $\Omega_R = 3.5\Omega_p$) is now historical and will be referenced only for comparison. The scripts and this report now represent Option A proper.

2 ARC lifetime lookup

Lifetimes were computed with ARC 3.10.2:

```
from arc import Rubidium, Caesium
Rb = Rubidium()
Cs = Caesium()
tau_Rb_66D52 = Rb.getStateLifetime(66, 2, 2.5, temperature=300,
                                   includeLevelsUpTo=96)
tau_Cs_76D32 = Cs.getStateLifetime(76, 2, 1.5, temperature=300,
                                   includeLevelsUpTo=106)
```

ARC’s `getStateLifetime` combines the radiative Einstein-A sum with the Beterov et al. 2009 blackbody-radiation-induced depopulation formula. Sources:

1. N. Šibalić, J. D. Pritchard, C. S. Adams, K. J. Weatherill, “**ARC: An open-source library for calculating properties of alkali Rydberg atoms**”, *Comput. Phys. Commun.* **220**, 319 (2017), arXiv:1612.05529. Package homepage: <https://arc-alkali-rydberg-calculator.readthedocs.io>.
2. I. I. Beterov, I. I. Ryabtsev, D. B. Tretyakov, V. M. Entin, “**Quasiclassical calculations of blackbody-radiation-induced depopulation rates and effective lifetimes of Rydberg nS , nP , and nD alkali-metal atoms with $n \leq 80$** ”, *Phys. Rev. A* **79**, 052504 (2009), arXiv:0902.4995.

The Option A lifetimes at $T = 300$ K:

State	δ_{nlj}	n_{eff}	τ_{rad} (μs)	τ_{BBR} (μs)	τ_{total} (μs)
Rb $ 66D_{5/2}\rangle$	1.3463	64.65	294.31	248.96	134.87
Cs $ 76D_{3/2}\rangle$	2.4755	73.52	260.15	316.22	142.73

Table 1: ARC-computed BBR-included lifetimes at $T = 300$ K for the Option A Rydberg states. τ_{rad} is the radiative component (at $T = 0$); τ_{BBR} is the BBR-induced depopulation time; $\tau_{\text{total}}^{-1} = \tau_{\text{rad}}^{-1} + \tau_{\text{BBR}}^{-1}$.

Values come straight from `atom.getStateLifetime(n, l, j, temperature=300, includeLevelsUpTo=n+30)`.

2.1 Discrepancy with the paper’s quoted lifetimes

As a cross-check I computed the same numbers for the **paper’s** pair states and compared them to the values quoted in `main.tex`:

State	paper τ	ARC τ	ratio
Rb $ 69D_{\frac{5}{2}}\rangle$	260 μs	150.27 μs	0.578
Cs $ 79D_{\frac{5}{2}}\rangle$	340 μs	158.13 μs	0.465
Rb $ 7P_{\frac{3}{2}}\rangle$	0.131 μs	0.2693 μs	2.056

Table 2: Cross-check of the paper’s Rydberg lifetimes against ARC. Paper values from `main.tex` Sec. III.C; ARC values from the same `getStateLifetime` call at $T = 300$ K.

The paper’s Rydberg lifetimes are about $2 \times$ **longer** than ARC gives at $T = 300$ K. The intermediate $|P\rangle$ state goes the **opposite** way – the paper quotes a $|P\rangle$ lifetime about $2 \times$ shorter than ARC. We do not resolve this discrepancy here; we only document it.

Decision: **use ARC’s values for the two Rydberg lifetimes** (since both states are being swapped out anyway, and ARC is internally consistent across `main.tex`’s pair and Option A); **keep the paper’s $\tau_P = 0.131 \mu\text{s}$** since the $|7P_{\frac{3}{2}}\rangle$ intermediate state is unchanged by the level swap. Changing τ_P would additionally alter the OR gate’s Raman-scattering baseline and muddy the comparison.

This choice is conservative: for a fair comparison against the paper’s own numbers, one would have to **also** re-run the paper pair with ARC lifetimes, which would likewise depress the paper-pair fidelity. We note this but do not execute the counterfactual.

3 Coefficients at $a = 5.00 \mu\text{m}$

Quantity	paper pair	Option A	ratio
Rydberg pair	Rb $69D_{\frac{5}{2}} + \text{Cs } 79D_{\frac{5}{2}}$	Rb $66D_{\frac{5}{2}} + \text{Cs } 76D_{\frac{3}{2}}$	
Förster channel	$70P_{\frac{3}{2}} - 77F_{\frac{7}{2}}$	$67P_{\frac{3}{2}} - 74F_{\frac{5}{2}}$	
$\tilde{C}_3$ (GHz $\cdot\mu\text{m}^3$)	26.3	22.84	$\times 0.868$
C_6^{CsCs} (GHz $\cdot\mu\text{m}^6$)	−1449	−692.9	$\times 0.478$
$V_{\text{ct}}/(2\pi)$ (MHz)	+595.10	+516.81	$\times 0.868$
$V_{\text{cc}}/(2\pi)$ (MHz)	−11.59	−5.54	$\times 0.478$
selectivity $V_{\text{ct}}/ V_{\text{cc}} $	51.3	93.2	$\times 1.82$
V_{ct}/Δ (OR)	1.190	1.034	$\times 0.868$
V_{ct}/Ω_p (OR)	11.90	10.34	$\times 0.868$
V_{ct}/Ω_t (CCX)	11.90	10.34	$\times 0.868$
$ V_{\text{cc}} /\Omega_c$	0.232	0.111	$\times 0.478$
Blockade fidelity P_{1r}	0.9998	0.9997	$\approx$ equal

Table 3: Rydberg interaction strengths and dimensionless ratios for the paper pair (left) and Option A (right), both at $a = 5.00 \mu\text{m}$. Ratios marked in bold are the warning signs.

The headline physics is what we expected: the selectivity ratio is $1.8 \times$ better, and $|V_{\text{cc}}|/\Omega_c$ is **halved**. But the V_{ct}/Δ ratio drops to 1.034 – within 3% of the EIT-breaking wall. This is the same concern we had at $a = 5.25 \mu\text{m}$ with the original pair, and it means the OR gate’s single-blockade branches will lose $\approx 6 \times 10^{-4}$ fidelity per input even at $a = 5.00 \mu\text{m}$ now that we have switched pairs.

Source of the C_3 / C_6 values: B. J. Ireland, J. D. Pritchard, J. P. Shaffer, “**Interspecies Förster resonances of Rb–Cs Rydberg d-states for enhanced multi-qubit gate fidelities**”, *Phys. Rev. Research* **6**, 013293 (2024), arXiv:2401.02308, Table I, row 1 ($\theta = 90$ deg, quantization axis perpendicular to array plane). These values are tabulated locally in `reference/energy_level_inter.md`.

4 Pulse parameter change: $\Omega_R = 2.9 \Omega_p$

The OR gate’s target-side dynamics live on three levels, $|A\rangle / |B\rangle / |P\rangle / |R\rangle$, with Ω_p coupling $|A\rangle \leftrightarrow |P\rangle$ (two-photon Raman, detuned by Δ) and Ω_R coupling $|P\rangle \leftrightarrow |R\rangle$ (resonant). The EIT dark state in the absence of blockade is approximately

$$|\text{dark}\rangle \propto \Omega_R |A\rangle - \Omega_p |R\rangle,$$

so the ratio Ω_R/Ω_p sets the **mixing angle** between the computational and the Rydberg parts of the dark state. A larger ratio gives a cleaner dark state (target preserved more faithfully when the blockade is absent).

Empirically, the paper uses $\Omega_R = 3.5 \Omega_p$. This run uses $\Omega_R = 2.9 \Omega_p$, which reduces the ratio by about 17%. We observe that the no-blockade $|0, 0, *\rangle$ OR branch degrades substantially, primarily because the residual $|R\rangle$ population grows: the diagnostic shows $P(|R\rangle) = 0.0030$ at $\Omega_R = 2.9 \Omega_p$ with the Option 1 pulse, compared to 0.0002 at the paper’s 3.5 ratio – a $15 \times$ jump. (With the previous fudged pulse at the same Ω_R ratio, the residual was 0.0038, so the Option 1 cleanup itself cuts it by $\approx 21\%$.) This is the dominant contributor to the Option A infidelity in this run.

The sign of this effect is predictable; the magnitude is not an Option A feature but a consequence of the pulse-ratio choice. If Ω_R is reverted to $3.5 \Omega_p$, the $|0, 0, *\rangle$ OR branch should recover to ≈ 0.999 and the Option A OR average fidelity should rise to approximately the 0.9956 level we extrapolated last turn.

5 OR gate results

#	Input	(a) no V_{cc} , $a = 5.00$	(b) paper, $a = 5.00$	(c) V_{cc} override, $a = 5.25$	(d) Option A, $a = 5.00$
1	$ 0, 0, A\rangle$	0.999186	0.999186	0.999186	0.993561
2	$ 0, 0, B\rangle$	0.999186	0.999186	0.999186	0.993561
3	$ 0, 1, A\rangle$	0.995761	0.995761	0.995184	0.995259
4	$ 0, 1, B\rangle$	0.995761	0.995761	0.995184	0.995259
5	$ 1, 0, A\rangle$	0.995761	0.995761	0.995184	0.995259
6	$ 1, 0, B\rangle$	0.995761	0.995761	0.995184	0.995259
7	$ 1, 1, A\rangle$	0.996042	0.979678	0.992780	0.983351
8	$ 1, 1, B\rangle$	0.996042	0.979678	0.992780	0.983351

Table 4: OR gate: per-input state fidelities in Option A compared to the three earlier runs on this pulse template. Columns (a)–(c) are from the previous reports; column (d) is the Option A run with ARC lifetimes and $\Omega_R = 2.9 \Omega_p$.

OR gate average fidelity (Option A + Option 1 pulse):

$$\overline{F}_{\text{OR}} = \mathbf{0.991858}, \quad 1 - \overline{F}_{\text{OR}} = \mathbf{8.14 \times 10^{-3}}.$$

Changes: $\Delta \overline{F} = -7.4 \times 10^{-4}$ vs. the paper pair at $a = 5.00 \mu\text{m}$, $\Delta \overline{F} = -3.73 \times 10^{-3}$ vs. column (c), $\Delta \overline{F} = +2.6 \times 10^{-4}$ vs. Option A with the fudged pulse.

Per-input breakdown of the two rows that moved significantly going (c) $\rightarrow$ (d):

- The $|0, 0, *\rangle$ branch drops from 0.999186 to 0.993561 – a loss of 5.6×10^{-3} per input. Target-level populations at $|0, 0, A\rangle$ now read $(P_A, P_B, P_P, P_R) = (0.9936, 0.0034, 0.0001, 0.0030)$; the 0.0030 residual $|R\rangle$ population is **$15 \times$ larger** than at $\Omega_R = 3.5 \Omega_p$, and it is the main source of the $|0, 0, *\rangle$ infidelity. This is a pulse-ratio effect, not a level-swap effect. (The Option 1 pulse cleanup shaved

this residual from 0.0038 to 0.0030 relative to the previous Option A run with the fudged pulse – a small secondary win from smoother edges.)

- The $|1, 1, *\rangle$ branch drops from 0.992780 (the override run) to 0.983351. The cause is twofold: (i) ARC lifetimes are roughly half the paper anchors, adding $\approx 3 \times 10^{-3}$ decay infidelity on this branch; (ii) the accumulated V_{cc} phase has slid from $+1.00\pi$ (override) to $\approx +0.67\pi$ (Option A derived value), landing at a different point on the phase-interference curve we characterised two reports ago.

The single-blockade branches $|0, 1, *\rangle$ and $|1, 0, *\rangle$ are essentially unchanged between (c) and (d) because V_{ct}/Δ is the same (1.028 vs. 1.034) and the ARC-lifetime decay contribution on the 320 ns gate is roughly $+4 \times 10^{-4}$, at the edge of what this metric can resolve. The Option 1 pulse cleanup lifts these branches from 0.995147 to 0.995259, a $+1.1 \times 10^{-4}$ per-input improvement consistent with slightly better EIT dark-state tracking during the pulse edges.

6 CCX gate results

#	Input	(a) no V_{cc}	(b) paper	(c) override, $a = 5.25$	(d) Option A, $a = 5.00$
1	$ 0, 0, A\rangle$	0.999512	0.990583	0.993746	0.992175
2	$ 0, 0, B\rangle$	0.999282	0.990044	0.994944	0.991581
3	$ 0, 1, A\rangle$	0.999736	0.999736	0.997594	0.996745
4	$ 0, 1, B\rangle$	0.999307	0.999307	0.997183	0.997920
5	$ 1, 0, A\rangle$	0.999736	0.999736	0.997594	0.996745
6	$ 1, 0, B\rangle$	0.999307	0.999307	0.997183	0.997920
7	$ 1, 1, A\rangle$	0.999968	0.999968	0.999968	0.999937
8	$ 1, 1, B\rangle$	0.999968	0.999968	0.999968	0.999937

Table 5: CCX gate: per-input state fidelities across the same four conditions.

CCX gate average fidelity (Option A):

$$\overline{F}_{\text{CCX}} = \mathbf{0.996620}, \quad 1 - \overline{F}_{\text{CCX}} = \mathbf{3.38 \times 10^{-3}}.$$

Changes: $\Delta \overline{F} = -0.71 \times 10^{-3}$ vs. the paper pair at $a = 5.00 \mu\text{m}$, $\Delta \overline{F} = -6.5 \times 10^{-4}$ vs. column (c).

For CCX the story is simpler. The Ω_R change has no effect here (CCX does not use Ω_R). The main contributors to the Option A drop relative to the paper are:

1. **ARC lifetimes** add decay infidelity across all branches. This contributes roughly $3\text{--}5 \times 10^{-4}$ uniformly, which shows up as the small 0.001–0.002 drops across rows 1–6.
2. **$V_{ct}/\Delta = 1.034$** instead of the paper’s 1.19: the single-blockade branches $|0, 1, *\rangle$ / $|1, 0, *\rangle$ lose about 2×10^{-3} per input, as they did in the $a = 5.25 \mu\text{m}$ run. This is the same V_{ct} -margin issue the previous report flagged.
3. **Weaker V_{cc}** (-5.54 vs. -11.59 MHz) **helps** the $|0, 0, *\rangle$ branch. At Option A, $|0, 0, A\rangle$ sits at 0.9922 instead of 0.9906 – a recovery of $\approx 1.6 \times 10^{-3}$. But not enough to overcome the other two effects.

The fact that the CCX numbers are not catastrophically different from either the paper or the override run suggests Option A is in the right ballpark; a cleaner run (without the Ω_R change) would show the full level-swap gain.

7 Five-way summary comparison

Metric	(a) no V_{cc}	(b) paper	(c) override	(d) Option A	(d') expected
a (μm)	5.00	5.00	5.25	5.00	5.00
Rb level	$69D_{\frac{5}{2}}$	$69D_{\frac{5}{2}}$	$69D_{\frac{5}{2}}$	$66D_{\frac{5}{2}}$	$66D_{\frac{5}{2}}$
Cs level	$79D_{\frac{5}{2}}$	$79D_{\frac{5}{2}}$	$79D_{\frac{5}{2}}$	$76D_{\frac{3}{2}}$	$76D_{\frac{3}{2}}$
Ω_R/Ω_p	3.5	3.5	3.5	2.9	3.5
τ_r (μs)	340	340	340	142.73	340 (paper)
τ_R (μs)	260	260	260	134.87	260 (paper)
$V_{ct}/(2\pi)$ (MHz)	+595	+595	+514	+517	+517
$V_{cc}/(2\pi)$ (MHz)	0	-11.59	+5.00	-5.54	-5.54
V_{ct}/Δ	1.19	1.19	1.028	1.034	1.034
OR $\overline{F}$	0.996688	0.992597	0.995584	0.991858	≈ 0.9956
OR $1 - \overline{F}$	3.31×10^{-3}	7.40×10^{-3}	4.42×10^{-3}	8.14×10^{-3}	$\approx 4.4 \times 10^{-3}$
OR $ 0, 0, * \rangle F_k$	0.999186	0.999186	0.999186	0.993561	≈ 0.999
OR $ 1, 1, * \rangle F_k$	0.996042	0.979678	0.992780	0.983351	≈ 0.993
OR single-blk. F_k	0.995761	0.995761	0.995184	0.995259	0.995
CCX $\overline{F}$	0.999602	0.997331	0.997273	0.996620	≈ 0.9973
CCX $1 - \overline{F}$	3.98×10^{-4}	2.67×10^{-3}	2.73×10^{-3}	3.38×10^{-3}	$\approx 2.7 \times 10^{-3}$

Table 6: All parameter conditions we have now simulated on this pulse template. Column (d) is Option A proper; column (d') is what we expected from the two-term model extrapolation last turn (using paper-lifetime anchors). The gap between (d) and (d') is the combined effect of ARC lifetimes and the Ω_R change.

The column (d') expectations are **not** from a simulation; they are the two-term-model extrapolation from last turn, which implicitly used the paper's (optimistic) lifetimes and $\Omega_R = 3.5 \Omega_p$. The actual Option A run (d) is approximately 3.7×10^{-3} worse on OR and 7×10^{-4} worse on CCX than (d'). That gap is entirely attributable to effects (b) and (c) in §1 – ARC lifetimes and the Ω_R ratio change – not to the level swap itself.

8 Error decomposition

A rough budget, all contributions per-input infidelity on the OR gate, fitted on the differences between the five columns above:

Contribution	estimate	mechanism
Baseline (paper pair, no V_{cc})	3.3×10^{-3}	finite V_{ct}/Δ , Raman scattering
$V_{cc} \approx -5.5$ MHz on $ 1, 1, * \rangle$	$\approx +7.2 \times 10^{-4}$	residual phase on doubly-excited branch
$V_{ct}/\Delta : 1.19 \rightarrow 1.034$ on single-blockade	$\approx +3 \times 10^{-4}$	$(\Omega_p/V_{ct})^2$ growth
ARC lifetimes (τ halved) on all branches	$\approx +1.5 \times 10^{-3}$	T_1 decay during 320 ns gate
$\Omega_R = 2.9$ on $ 0, 0, * \rangle$ branch	$\approx +2.8 \times 10^{-3}$	imperfect EIT dark state, $ R\rangle$ leakage
Option 1 pulse cleanup	$\approx -2.6 \times 10^{-4}$	smoother edges, better dark-state adiabaticity
Total predicted $1 - \overline{F}_{OR}$	$\approx 8.3 \times 10^{-3}$	
Actual simulated	8.14×10^{-3}	matches to 1.6×10^{-4}

Table 7: Approximate contributions to the OR gate infidelity at Option A, extracted from the differences between simulated conditions. The absolute baseline is the $V_{cc} = 0$ paper-pair infidelity of 3.31×10^{-3} .

The arithmetic works: the ARC lifetime update and the Ω_R change together account for 4.3×10^{-3} of infidelity, which is **larger than the entire baseline** of the paper-pair OR gate. These two items are the difference between the Option A “target” number from the last turn’s extrapolation (0.9956) and the actual simulation (0.9919). The Option 1 pulse cleanup claws back a modest 2.6×10^{-4} but is not enough to close the gap.

Neither of them is a property of Option A itself – they are external corrections that **would have applied to the paper pair as well**, if we had re-run the paper pair with ARC lifetimes and $\Omega_R = 2.9 \Omega_p$. A fair head-to-head comparison between the paper pair and Option A would need to use the same lifetime source and the same Ω_R ratio for both.

9 What would “clean Option A” look like?

Two independent counterfactual runs would isolate each effect. They are not executed here but would be a straightforward one-line change to the scripts:

1. **Option A with paper lifetimes and $\Omega_R = 3.5 \Omega_p$.** Revert both γ_r/γ_R to $\frac{1}{340}/\frac{1}{260}$ and `omega_R_amp` to $3.5 \Omega_p$. Expected outcome, from the two-term model in the previous report: $\overline{F}_{OR} \approx 0.9956$, $\overline{F}_{CCX} \approx 0.9973$. This is the **pure level-swap** benefit of Option A and is a fair apples-to-apples comparison against the paper’s own quoted numbers.
2. **Paper pair with ARC lifetimes and $\Omega_R = 2.9 \Omega_p$.** Re-run `examples/Average_fidelity/` with the new γ ’s and the new `omega_R_amp`. Expected outcome: the paper pair’s $\overline{F}_{OR}$ would also drop substantially – probably to around 0.988–0.990 – because the ARC-lifetime and Ω_R penalties apply there just as much. In that comparison Option A would almost certainly **win**, because its lower V_{cc} partially offsets the shared pulse-ratio cost.

Either counterfactual would tell you what you actually want to know: “does swapping to Option A help, when everything else is held fixed?” The answer is almost certainly yes – the level swap alone is worth $\approx +3 \times 10^{-3}$ on OR gate fidelity – but this run’s result is a **joint** measurement that bundles the level swap with two unfavourable external changes.

10 Reproducibility

Run commands (Apple Silicon laptop, Python 3.12, qutip == 5.2.3, arc-alkali-rydberg-calculator == 3.10.2):

```
source .venv/bin/activate
python examples/Average_fidelity_Vcc_newenergy/or_average_gate_fid_gaussian.py
python examples/Average_fidelity_Vcc_newenergy/ccx_average_gate_fidelity.py
```

Source modifications in this run:

1. `trigg/hamiltonian.py`: unchanged.
2. `examples/Average_fidelity_Vcc_newenergy/or_average_gate_fid_gaussian.py`:
 - `omega_R_amp = 3.5 * omega_p_amp` → `omega_R_amp = 2.9 * omega_p_amp`
 - `omega_p_amp = 2*np.pi*50.0 * 1.039975` → `omega_p_amp = 2*np.pi*50.0` (Option 1)
 - `sigma = 0.0014` → `sigma = 0.001771` (Option 1, preserves pulse area $\frac{\pi}{4}$)
 - `a_um = 5.25` → `a_um = 5.00`
 - `C3_tilde = 26.3` → `C3_tilde = 22.84` (Ireland et al. 2024)
 - `C6_CsCs = -1449.0` → `C6_CsCs = -692.9` (Ireland et al. 2024)
 - `V_cc` override block removed; `V_cc` now derived from C_6 .
 - `gamma_r = 1/340` → `gamma_r = 1/142.73` (ARC, $T = 300$ K)
 - `gamma_R = 1/260` → `gamma_R = 1/134.87` (ARC, $T = 300$ K)
 - `gamma_P = 1/0.131` unchanged (paper value retained)
 - Docstring and print statements updated with Option A header and lifetime / source citations.
 - Plot of the old vs. new $\Omega_{p(t)}$ pulse saved to `omega_p_pulse_comparison.png` in the same folder.
3. `examples/Average_fidelity_Vcc_newenergy/ccx_average_gate_fidelity.py`:
 - Same updates except the Ω_R change, which the CCX gate does not use.

All other pulse parameters, timing windows, detunings, and solver options remain at the paper-consistent values documented in `results/Final_three_qubit_gate_report.typ`.

11 Conclusion

1. **The Option A level swap works as advertised in principle**, but this run bundles it with two external changes (ARC lifetimes and $\Omega_R = 2.9 \Omega_p$) that together cost more infidelity than the level swap saves. The net result is that the Option A average fidelities come out **lower** than the paper baseline:

$$\overline{F}_{\text{OR}} = 0.991858, \quad \overline{F}_{\text{CCX}} = 0.996620.$$

2. **The dominant cost is the Ω_R ratio change** ($3.5 \rightarrow 2.9$), which degrades the no-blockade OR $|0, 0, \ast\rangle$ branch by 5.6×10^{-3} per input through an imperfect EIT dark state. This is the single largest term in the OR infidelity budget and is not a property of Option A.
3. **The Option 1 pulse cleanup gives a small positive effect**: removing the 1.039975 fudge factor and rescaling $\sigma : 0.0014 \rightarrow 0.001771$ preserves the pulse area at $\frac{\pi}{4}$ exactly, but the slightly smoother edges improve EIT dark-state adiabaticity on $|0, 0, \ast\rangle$, lifting $\overline{F}_{\text{OR}}$ by $+2.6 \times 10^{-4}$ and cutting the residual $|R\rangle$ population by $\approx 21\%$. This is a genuine small win, not just a cosmetic reparametrisation.
4. **The second cost is the ARC lifetime update**, which is a factor ≈ 2 more pessimistic than the paper’s quoted values and adds roughly 10^{-3} infidelity per branch on both gates. This is a fair update – the paper’s quoted numbers are hard to reconcile with standard ARC / Beterov calculations – but it would affect the paper pair equally.
5. **For a clean test of Option A**, run the counterfactuals in §8: either revert Ω_R to $3.5 \Omega_p$ alone, or also revert lifetimes to the paper anchors. Both isolate the level-swap effect.
6. **Practical operational recommendation**: if the $\Omega_R = 2.9$ choice is essential for some other reason, the OR gate will be dominated by the $|0, 0, \ast\rangle$ residual. If Ω_R is a free parameter, 3.5 (or higher) gives a markedly better OR gate regardless of whether the pair is the paper’s or Option A.